

# Topic 13: Variation of Parameters & Cauchy-Euler Equations

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## Professor Leonard Master Series · Variation of Parameters

**Course:** Applied Ordinary Differential Equations (ENGR 213 Companion)

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### 1. Whiteboard Intuition: When Guessing Fails

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What happens if the driving function  $g(x)$  is  $\tan(x)$ ,  $\sec(x)$ ,  $\frac{1}{x}$ , or  $\ln(x)$ ? Derivatives of  $\tan(x)$  produce  $\sec^2(x)$ ,  $2\sec^2(x)\tan(x)$ ,  $\dots$  — an infinite cascade of new functions! The Method of Undetermined Coefficients completely fails.

We need a **universal method: Variation of Parameters**. Instead of constant parameters  $C_1, C_2$ , let them vary as functions:

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

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### 2. The Universal Wronskian Formula

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For  $y'' + P(x)y' + Q(x)y = f(x)$ :

$$W = \begin{vmatrix} y_1 & y_2 & y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2$$
$$u_1(x) = - \int \frac{y_2(x)f(x)}{W} dx, \quad u_2(x) = \int \frac{y_1(x)f(x)}{W} dx$$

Then:

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

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### 3. Cauchy-Euler Equations: $ax^2y'' + bxy' + cy = 0$

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Notice that each power of  $x$  matches the order of the derivative ( $x^2y''$ ,  $xy'$ ,  $x^0y$ ).

- Substitute  $y = x^m \implies y' = mx^{m-1} \implies y'' = m(m-1)x^{m-2}$ .
- Auxiliary equation:

$$am(m-1) + bm + c = 0 \iff am^2 + (b-a)m + c = 0$$

- **Three Cases:**

1. Distinct real  $m_1 \neq m_2$ :  $y = C_1x^{m_1} + C_2x^{m_2}$
2. Repeated real  $m_1 = m_2 = m$ :  $y = C_1x^m + C_2x^m \ln x$
3. Complex  $m = \alpha \pm i\beta$ :  $y = x^\alpha (C_1 \cos(\beta \ln x) + C_2 \sin(\beta \ln x))$